

Risk	Probability	Impact	Score	Priority	Response Strategy	Owner	Monitoring plan
Access delays to demo robot logs/APIs	2	3	6	Critical	Mitigation (secure credentials early and pre-provision VPN/network), Contingency (run a mock data generator)	Data Engineer	Check access request status weekly; flag if no progress >3 business days; try a daily connection to confirm the system is reachable.
Poor/unstable data quality	3	2	6	Critical	Mitigation (add schema/timestamp validation, resampling, and minimum data-quality gates before modeling), Monitoring (daily data-quality checks)	Data Scientist/Data Engineer	Daily data check: missing timestamps, irregular timing, offline sensors; alert when limits are crossed.
Confidentiality/privacy constraints	2	2	4	Moderate	Mitigation (use read-only access, anonymize IDs, store in university Azure tenancy), Acceptance (tolerate minor collaboration friction)	Data Engineer	Monthly review of who has access; auto-check data is in the university cloud; confirm IDs are anonymized in the data flow.
SAM Automation availability gaps	3	2	6	Critical	Mitigation (schedule weekly 30-min review and appoint a backup with sign-off), Monitoring (alert on >1 missed session)	Scrum Master	Track expert meeting attendance and decision time; alert after >1 missed meeting or >3 days to decide.
Technician resistance to change	2	2	4	Moderate	Mitigation (co-design top 2–3 alerts; add “snooze” and “why this alert?”), Monitoring (track adoption and dismiss rates)	Data Scientist/Analytics Translator	Track how often alerts are dismissed or snoozed and time-to-action; flag if <70% of alerts lead to action; add a short user survey.
Changing requirements	2	2	4	Moderate	Mitigation (lock scope after week 4, log any change requests, move non-essential changes to phase 2), Monitoring	Scrum Master	Count weekly change requests; if >2 must-have changes arrive in a week, hold a scope review.
Insufficient failure labels	3	2	6	Critical	Mitigation (start with threshold/anomaly baselines and SME rules), Contingency (log outcomes to build future supervised labels)	Analytics Translator	Record the share of alerts experts review and label; aim ≥60% by Week 14 (tbd); review incorrect alerts weekly.
Tooling/IT constraints	2	2	4	Moderate	Mitigation (decide toolset early - Streamlit or Power BI and package dashboard & helper services in Docker/OCI with pinned versions), Contingency (provide a no-install demo image and runbook)	Data Scientist/Data Engineer	Weekly quick test of the packaged app on lab computers; track support issues and library/version changes; verify the step-by-step guide works.
Local vs cloud processing mismatch	2	2	4	Moderate	Mitigation (pilot hybrid: on-robot summaries → central store; make sampling/config refresh configurable), Monitoring (track latency, bandwidth, and CPU)	Data Engineer	Measure response time, network use, processor load, and data loss; alert if slow responses exceed 500 ms or data loss >1%.
Scope creep toward production	2	3	6	Critical	Avoidance (reiterate “prototype only”; put production requests on a later list), Monitoring (block items that need service guarantees or big system hookups)	Analytics Translator	Count requests that look like full production work; stop them in reviews; alert if any enter the current work plan.
Team availability conflicts	2	2	4	Moderate	Mitigation (share a team calendar, cross-train team members, do must-haves first), Contingency (use a backup person if someone is out)	Scrum Master	Keep a calendar of planned absences; alert when a task owner will be away during a critical task; ensure a backup is named.
Unexpected data/ETL complexity	3	2	6	Critical	Mitigation (run a short Week 5-6 data-processing test, standardize data formats, focus on the 3 most reliable signals), Contingency (cut back on charts if needed)	Data Engineer	Track the share of files that parse correctly, the number of retries, and data-format changes; alert if success <90% for two days.
Vendor/API changes (firmware/API updates)	2	2	4	Moderate	Mitigation (pin firmware/API versions and maintain a test harness), Contingency (keep a mock feed), Monitoring (subscribe to vendor release notes)	Data Engineer	Read vendor release notes; run weekly compatibility tests; alert if versions change or tests fail.